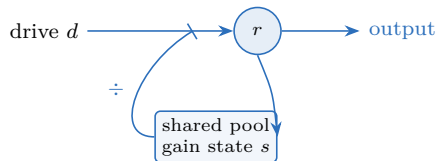


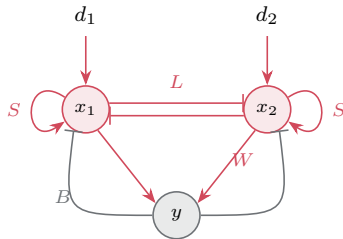
**a** divisive normalization (DN)



gain state  $s \in \{0, \dots, 3\}$

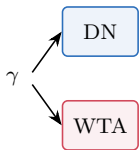
$$s' = \text{round}\left(2d/(2+s)\right), \text{ clipped}$$

**b** winner-take-all competition (WTA)

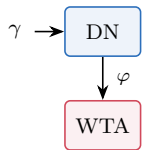


circuit state  $(x_1, x_2, y) \in \{0, 1\}^3$ :  
winners and inhibitory gate

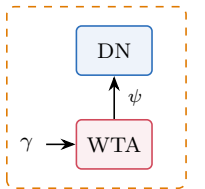
**c** compositions on the product state space  $X_D \times X_W$



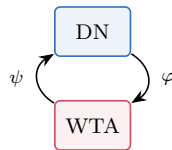
independent  
product



DN→WTA  
cascade



WTA→DN cascade  
(headline)



recurrent  
(synchronous or sequential)

interfaces:  $\varphi$ : DN gain state  $\mapsto$  WTA drive condition;  $\psi$ : winner & gate  $\mapsto$  DN drive symbol (winner pooling); recurrent updates take no external symbol